Sequences & Series

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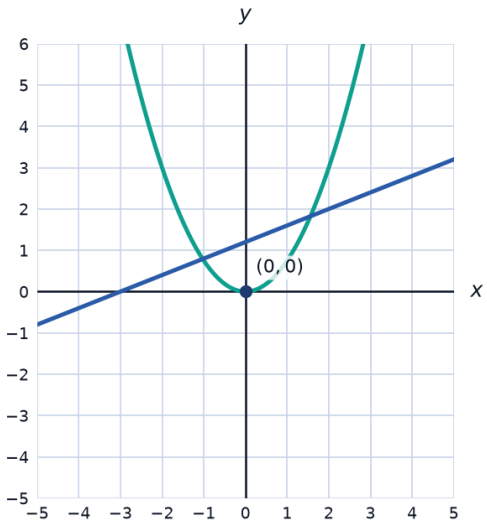
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

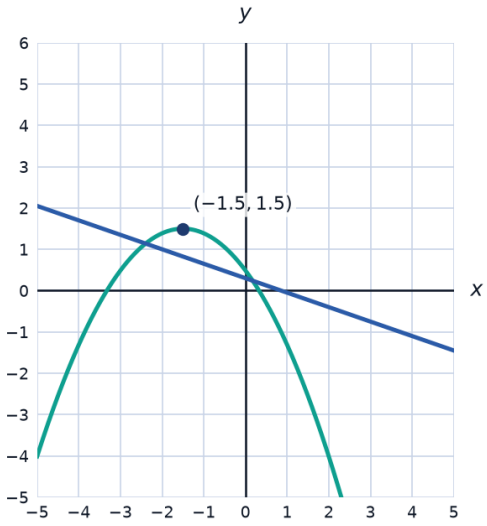
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

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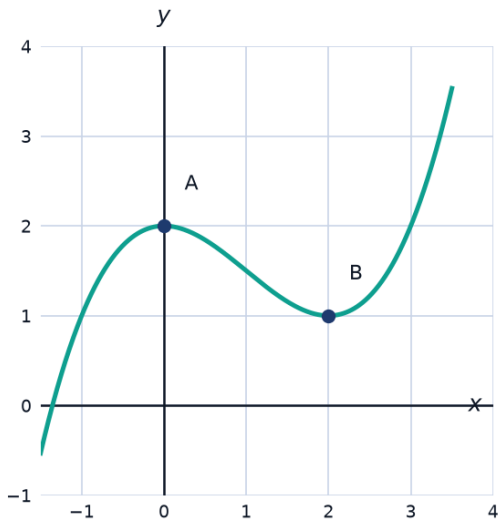
6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

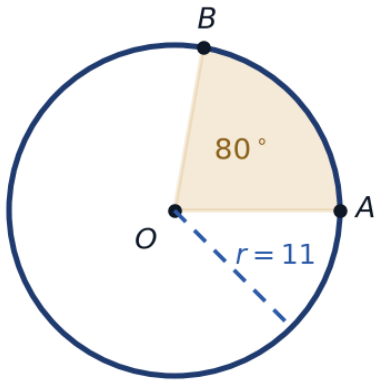
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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2 + y^2}\, dA$ where D is the semi-annular region $\{(x, y) : 1 \le x^2 + y^2 \le 4, y \ge 0\}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi} \int_y^{\pi} \frac{\sin x}{x}\, dx\, dy$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

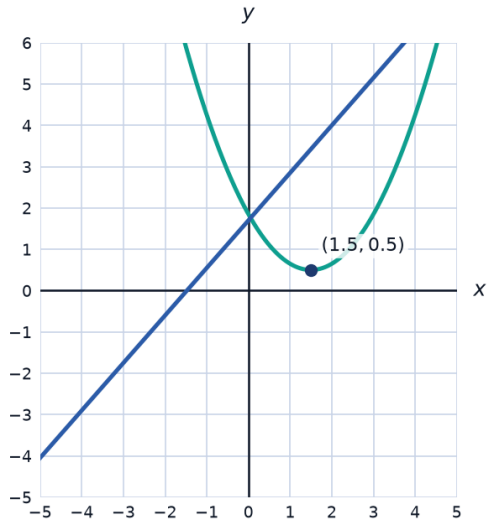
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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Vectors

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Complex Numbers

Find the Maclaurin series for $\ln(1 + 2x)$ up to the x^3 term, and state its interval of convergence.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Numerical Methods

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Differential Equations

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first three terms in the expansion of $(1 + 4x)^{1/2}$ in ascending powers of x.

a. $1 + 2x - 2x^2$

b. $1 + 2x + 2x^2$

c. $1 + 4x - 8x^2$

d. $1 + 2x - 4x^2$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the range of values of x for which the binomial expansion of $(1 + 4x)^{1/2}$ is valid.

a. $|x| < \frac{1}{4}$

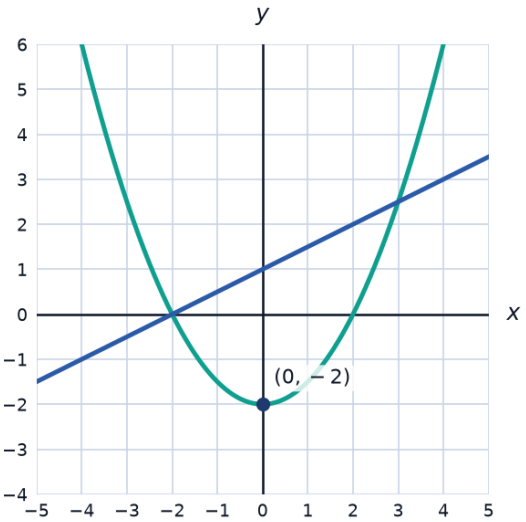
b. $|x| < 4$

c. $|x| < 1$

d. $x > 0$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Expand $(1 - 2x)^{-1}$ up to the term in x^3.



a. $1 + 2x + 4x^2 + 8x^3$

b. $1 - 2x + 4x^2 - 8x^3$

c. $1 + 2x + 2x^2 + 2x^3$

d. $1 - 2x - 4x^2 - 8x^3$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use the first three terms of $(1 - x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

a. 0.98995

b. 0.98990

c. 0.99000

d. 0.98500

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

a. $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$

b. $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

c. $(50)^{1/2} = (1+49)^{1/2}$

d. $\sqrt{36+14} = 6(1+14/36)^{1/2}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coefficient of x^2 in the expansion of $\frac{1+2x}{(1-x)^2}$.

a. 5

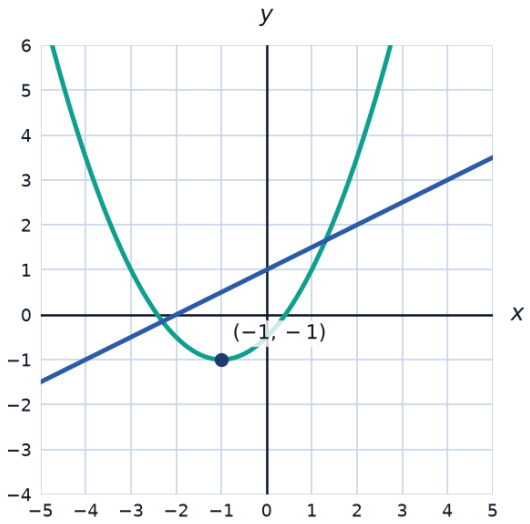
b. 3

c. 4

d. 6

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.



a. $\frac{n}{n+1}$

b. $1-\frac{1}{n}$

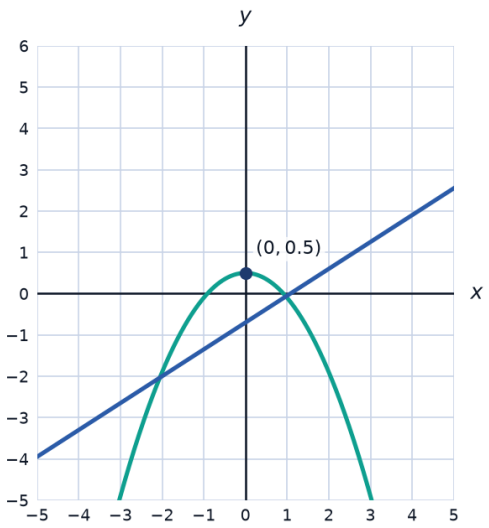
c. $\frac{n+1}{n}$

d. $1-\frac{1}{(n+1)^2}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum to infinity $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.



a. 1

b. 2

c. 1/2

d. ∞

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

a. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$

b. $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

c. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

d. $f(x) = f(1) + f'(1)(x-1) + \ldots$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

a. $x - \frac{x^2}{2} + \frac{x^3}{3}$

b. $x + \frac{x^2}{2} + \frac{x^3}{3}$

c. $x - x^2 + x^3$

d. $1 + x - \frac{x^2}{2}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term $a = 2$ and common difference $d = -5$. Find the 8th term.

a. -33

b. -35

c. -38

d. 16

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $x^2 + 10x + 41 = 0$. Find the sum of the roots.

a. 41

b. -9

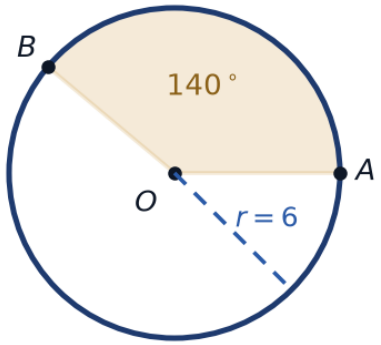
c. 10

d. -10

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The locus $|z-(-6+i)|=|z-(2-2i)|$ is the perpendicular bisector of the segment joining the points representing z_1 and z_2. Find the x-coordinate of the midpoint of this segment.



a. $-\frac{1}{2}$

b. 2

c. -2

d. -6

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the sequence defined by $a_1=1$ and $a_{n+1}=\frac{a_n^2+k}{2}$ for $n\geq 1$, where k is a positive constant. If the sequence converges to a limit L, prove that $L^2-2L+k=0$. Furthermore, if the limit L is an integer and $k=3$, determine the exact value of the limit.

a. L = 3

b. L = 1

c. L = -1

d. $L=\sqrt{3}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sequence is defined by $a_1=1$ and $a_{n+1}=a_n+2n+1$ for $n\geq 1$. Prove by induction that $a_n=n^2$ for all positive integers n. Then, consider the series $S_N=\sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10}(a_n-a_{n-1})$ where $a_0=0$.

a. 100

b. 110

c. 10

d. 210

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

c. 6

d. 7

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. 2

b. 0

c. -2

d. 1

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Numerical Methods

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

a. 6

b. 7

c. 8

d. 9

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Differential Equations

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the 4th term of this sequence, a_4?

a. 11

b. 14

c. 17

d. 20

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Poisson Distribution

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fourth term of this sequence, a_4?

a. 14

b. 11

c. 17

d. 8

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 13.01 Continuous Random Variables

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?

a. 17

b. 20

c. 15

d. 23

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 14.01 Sampling & Estimation

Product of two power series $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:

a. $a_n b_n$

b. $a_n + b_n$

c. $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product)

d. $a_0 b_n$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Hypothesis Testing

Integral test: $\sum_{n=1}^{\infty} f(n)$ converges iff $\int_{1}^{\infty} f(x)\,dx$ converges, when f is:

a. any function

b. bounded

c. increasing

d. positive, decreasing, continuous

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Exponentials & Logarithms

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge absolutely?

a. Yes

b. Yes, to $-\ln 2$

c. No ($\sum 1/n$ diverges)

d. Inconclusive

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